

Hartshorne Chapter 2 Section 1: Sheaves

Zih-Yu Hsieh

February 19, 2026

1 Presheaves and Sheaves

Definition: Presheaves

Given a topological space X , let $\mathbf{Top}(X)$ denotes the category of partial orders on open subsets of X , with morphisms being inclusions.

A presheaf F of abelian groups on X is a functor $F : \mathbf{Top}(X)^{\text{op}} \rightarrow \mathbf{Ab}$, where:

1. For any inclusion of open subsets $V \hookrightarrow U$, there exists a „restriction“ group homomorphism $\rho_{UV} : F(U) \rightarrow F(V)$.
2. The morphism satisfies $\rho_{UU} = \text{id}_{F(U)}$, and given open subsets $W \hookrightarrow V \hookrightarrow U$, one has $\rho_{UW} = \rho_{VW} \circ \rho_{UV}$.

Sometimes there is an extra restriction $F(\emptyset) = 0$

For convenience, given any $s \in F(U)$, and say $V \hookrightarrow U$, $\rho_{UV}(s)$ will be denoted as $s|_V$ (or s restricted to V). And, the group $F(U)$ will be denoted by $\Gamma(U, F)$.

Definition: Sheaves

Given a presheaf F on a topological space, it's called a *Sheaf* if it satisfies the following two axioms:

1. **Separation Axiom:** Given U an open set, with an open cover $\{U_i\}$, if $s \in F(U)$ satisfies $s|_{U_i} = 0$ for all index i , then $s = 0$.
2. **Gluing Axioms:** If U is an open set, with an open cover $\{U_i\}$. If each index i has an associated $s_i \in F(U_i)$, such that every index pair i, j satisfies $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$, then there exists an $s \in F(U)$, such that $s|_{U_i} = s_i$ for all index i .

Which, the Separation Axiom guarantees the uniqueness of the gluing in the Gluing Axiom: Suppose $s, s' \in F(U)$ are both gluings of $\{s_i \in U_i\}$, then one has $s|_{U_i} = s'|_{U_i} = s_i$. Using the group homomorphism property, $0 = s_i - s_i = s|_{U_i} - s'|_{U_i} = (s - s')|_{U_i}$ for all index i , hence separation axiom guarantees $s - s' = 0$, or $s = s'$ using the Abelian Group Property.

If given these two axioms, one must have $F(\emptyset) = 0$: Given any two elements $s, s' \in F(\emptyset)$ notice that they agree on $\emptyset \cap \emptyset = \emptyset$, so there must exist $s'' \in F(\emptyset)$, such that $s''|_{\emptyset} = s' = s''$, showing there must be only one element, namely 0. So, the condition that $F(\emptyset) = 0$ technically can be omitted from the presheaf definition.

Example: Classical Sheaves

Given a topological space X , smooth real manifold M , and complex manifold H . Let $U : \mathbf{Top}(X)^{\text{op}} \rightarrow \mathbf{Ab}$ denotes $U(V)$ as all continuous functions $V \rightarrow \mathbb{R}$, let $C^\infty : \mathbf{Top}(M)^{\text{op}} \rightarrow \mathbf{Ab}$ denotes $C^\infty(V)$ as all smooth functions $V \rightarrow \mathbb{R}$, and $\mathcal{O} : \mathbf{Top}(H)^{\text{op}} \rightarrow \mathbb{C}$ be all the holomorphic functions $H \rightarrow \mathbb{C}$. Also, the restriction map is the classical restriction of domains.

Then, by the gluing lemma, all of these satisfied gluing conditions; and, if a function is locally 0 every, it's 0 globally (on a given open subset). Hence, these are sheaves.

Example: Regular Functions on Varieties

Given X a variety over a field $k = \bar{k}$, let $\mathcal{O} : \mathbf{Top}(X)^{\text{op}} \rightarrow \mathbf{Ab}$ satisfies $\mathcal{O}(U)$ being a collection of regular functions $U \rightarrow k$, together with restrictions being the restriction of domains.

It's a sheaf, because a function who's locally 0 everywhere is 0 globally (on an open set); also, if the regular functions on open subsets agree on their overlap, then they glue o a regular function.

Example: Constant Presheaves & Sheaves

Fix $A \in \mathbf{Ab}$ a nonzero abelian group. The **Constant Presheaf** $A : \mathbf{Top}(X)^{\text{op}} \rightarrow \mathbf{Ab}$ on X satisfies $A(U) = A$ for all open subset $U \subseteq X$, and all inclusions are mapped to identity.

This is a presheaf (since identity vacuously composed to identity), but not a sheaf, since it no longer has the property $F(\emptyset) = 0$.

As a modification, given A a discrete topology, and let $\mathcal{A} : \mathbf{Top}(X)^{\text{op}} \rightarrow \mathbf{Ab}$ be a presheaf of X , such that $\mathcal{A}(U) = \text{Hom}_{\mathbf{Top}}(U, A)$, where open subset $U \subseteq X$ induces the subspace topology, while A maintains the discrete topology. Also, the restriction maps are by restriction of domains.

This is well-defined, as given A discrete topology, its addition $A \times A \rightarrow A$ is clearly continuous, which induces a pointwise addition structure on $\text{Hom}_{\mathbf{Top}}(U, A)$ by $(f + g)(a) = f(a) + g(a)$ (a continuous map by composing $(f, g) : U \rightarrow A \times A$ and $+ : A \times A \rightarrow A$).

Also, since each set are continuous maps, then the gluing axioms (together with uniqueness) both are satisfied here, hence it's a sheaf.

Finally, the reason it's called „constant sheaf“, is by the fact that if U is connected, then $\text{Hom}_{\mathbf{Top}}(U, A) \cong A$ (since all $a \in A$ defines a connected component, so all continuous function must be constant since it sends U connected to a connected subset, which can only be singleton). So, given any V with c being its collection of connected component, one has $\text{Hom}_{\mathbf{Top}}(V, A) \cong V^{\times c}$, the direct product with a copy for each connected component of V .

2 Stalks and Direct Limits

As a review, let's talk about direct limits:

Definition: Direct System

Given $(I, \leq)$ a preordered set, it satisfies:

1. **Reflexivity:** $a \leq a$ for all a
2. **Transitivity:** $a \leq b$ and $b \leq c$ implies $a \leq c$

Yet, it doesn't require antisymmetry, so $a \leq b$ and $b \leq a$ doesn't imply $a = b$; preorder is different from partial order because the lack of antisymmetry).

Now, $(I, \leq)$ is called a *Direct System*, if any finite set has an upper bound. Equivalently, given any $i, j \in I$, there exists $k \in I$ satisfying $i, j \leq k$.

Note that it forms a category due to Reflexivity.

Definition: Direct System in Category

A direct system in a category $\mathcal{C}$, is given by a direct system $(I, \leq)$ and a functor $F : I \rightarrow \mathcal{C}$.

Hence, for any $i \leq j$, it corresponds to a morphism $f_{ij} : F(i) \rightarrow F(j)$.

Given $i \leq j$ and $j \leq k$, one has $i \leq k$, where they correspond to morphisms $f_{ij} : F(i) \rightarrow F(j)$, $f_{jk} : F(j) \rightarrow F(k)$, and $f_{ik} : F(i) \rightarrow F(k)$. Which, transitivity states $f_{ik} = f_{jk} \circ f_{ij}$.

And, given any $i, j \in I$, there exists k , such that $i, j \leq k$ translates to the existence of morphisms $f_{ik} : F(i) \rightarrow F(k)$ and $f_{jk} : F(j) \rightarrow F(k)$ for some $k \in I$.

Definition: Direct Limit in Category

Given a category $\mathcal{C}$ with a direct system $F : I \rightarrow \mathcal{C}$, a *Direct Limit*, denoted as $\lim_{i \in I} F$, is formally defined as $\text{colim } F$.

As an example, let's take $\mathbf{Ab}$, a category where all limits / colimits exist:

Example: Direct Limit in $\mathbf{Ab}$

Let I be a direct system, a direct system in $\mathbf{Ab}$ based on I , is a collection of abelian groups $\{A_i\}_{i \in I}$, where $i \leq j$ corresponds to a group homomorphism $f_{ij} : A_i \rightarrow A_j$, and any i, j has a corresponding $k \in I$, where there exists group homomorphisms f_{ik}, f_{jk} .

Then, the direct limit is the group $A := \bigoplus_{i \in I} A_i / \sim$, where $\sim$ is the relation satisfying $a_i \sim a_j$ for $a_i \in A_i$ and $a_j \in A_j$, if there exists $k \in I$ with f_{ik}, f_{jk} being well-defined, such that $f_{ik}(a_i) = f_{jk}(a_j)$.

This is an equivalence relation, hence form a well-defined subgroup:

1. All $f_{ii}(a_i) = f_{ii}(a_i)$, showing any $a_i \in A_i$ satisfies $a_i \sim a_i$, proving reflexivity.
2. If $a_i \sim a_j$, there exists $k \in I$ with $f_{ik}(a_i) = f_{jk}(a_j)$, reversing the equality provides $a_j \sim a_i$.
3. If $a_i \sim a_j$ and $a_j \sim a_k$, there exists $l_1, l_2 \in I$, where $f_{il_1}(a_i) = f_{jl_1}(a_j)$, and $f_{jl_2}(a_j) = f_{kl_2}(a_k)$. Now, find an upper bound of l_1, l_2 , say $r \in I$. Then, one has the following:

$$f_{ir}(a_i) = f_{l_1 r} \circ f_{il_1}(a_i) = f_{l_1 r} \circ f_{jl_1}(a_j) = f_{jr}(a_j) = f_{l_2 r} \circ f_{jl_2}(a_j) = f_{l_2 r} \circ f_{kl_2}(a_k) = f_{kr}(a_k)$$

So, $a_i \sim a_k$, prove the transitivity.

Finally, the map $A_i \rightarrow A$ is defined by the composition of inclusion $A_i \hookrightarrow \bigoplus_{i \in I} A_i$ and projection $\bigoplus_{i \in I} A_i \twoheadrightarrow A$.

To show it's indeed a colimit of the given functor, suppose B is another abelian group with group homomorphisms $g_i : A_i \rightarrow B$ that forms a cocone of the given functor